

Course Code	Course Title	Semester	Category	Periods/Week				C
				L	T	P	R	
2321CSS301J	Embedded System Design	III	ES	3	0	2	0	4
Pre-requisite	Nil	Course Type	THEORY CUM PRACTICAL					

Course Objectives: This course will help the students

Understand the fundamentals of embedded systems .

Learn the communication protocols used in embedded systems.

Acquire programming skills in Embedded C for embedded systems development.

Understand RTOS concepts, task scheduling, synchronization mechanisms, and Edge AI/TinyML fundamentals for embedded applications.

Understand the embedded and Industrial IoT systems using FPGA-based acceleration and modern embedded technologies.

Course Outcomes:

Upon completion of the course, the students will be able to:

CO Number	Course Outcomes
CO1	Analyze the architecture, components of embedded systems and IoT-enabled devices.
CO2	Analyze communication protocols and interfacing techniques used in modern embedded applications.
CO3	Develop Embedded C programs using timers, interrupts, and multitasking concepts for real-time applications.
CO4	Apply RTOS concepts and Edge AI/TinyML frameworks in embedded system design.
CO5	Design Industrial IoT solutions incorporating FPGA-based acceleration and intelligent edge computing applications.

Course Content:

Unit I	Introduction to Embedded Systems	9
Definition and characteristics of embedded systems – Embedded vs general-purpose systems – Components of embedded systems – Types of embedded systems, Introduction to IoT-enabled embedded devices.		
Unit II	Embedded System Architecture	9
Hardware architecture: Von Neumann and Harvard, Microcontroller vs Microprocessor – Memory types and interfacing – I/O devices and interfacing – Bus systems and protocols (I2C, SPI, UART, CAN) – Introduction to FPGA architecture – Basics of hardware acceleration and parallel processing concepts.		
Unit III	Embedded C and Programming Concepts	9
Basics of Embedded C – I/O programming using C – Bit manipulation and memory access – Timer and interrupt programming – Introduction to multithreading concepts – Basics of parallel processing in embedded systems – Embedded software debugging techniques - Breakpoint, JTAG and SWD Debugging.		
Unit IV	RTOS and Task Management	9
Introduction to Real-Time Operating Systems (RTOS) – Scheduling algorithms: Round Robin, Priority Scheduling – Task creation and management – Synchronization: Semaphores, Mutexes – Inter-process communication – Introduction to Edge AI and TinyML concepts – Edge AI frameworks: TensorFlow Lite for Microcontrollers (TFLite Micro), TinyML basics.		
Unit V	Embedded System Design and Applications	9
Design life cycle and modeling – Embedded system testing and validation – Industrial IoT (IIoT) frameworks and architecture – FPGA-based embedded applications – Hardware acceleration techniques – Case studies: Smart healthcare, Industrial IoT devices and Edge AI applications.		

Platform Needed:			
Total Periods			45
List of Experiments			
S.No.	Title of the Experiments		
1	Applications of Timers and Interrupts in Microcontroller Programming		
2	Temperature Monitoring and Sensor Interfacing		
3	UART-based Serial Communication System		
4	RTOS-based Task Scheduling		
5	Case Studies: Smart Street Light / Home Automation System		
6	Design and Implementation of TinyML for IoT Applications		
Total Periods			30
Learning Resources:			
Text Books			
1	Raj Kamal , “Embedded Systems: Architecture, Programming and Design” , 3rd Edition, McGraw Hill		
2	Frank Vahid and Tony Givargis, “Embedded System Design: A Unified Hardware/Software Introduction”, Wiley		
3	Wayne Wolf – Computers as Components: Principles of Embedded Computing System Design, Morgan Kaufmann.		
Reference Books			
1	James K. Peckol, “Embedded Systems: A Contemporary Design Tool” , Wiley		
2	Lyla B.Das ,” Embedded Systems an Integrated Approach”, Pearson Education.		
Online Resources (Web Links)			
1			
2			
Name of the Internal Expert 1		Signature	
Name of the Internal Expert 2		Signature	